

ADVANCED

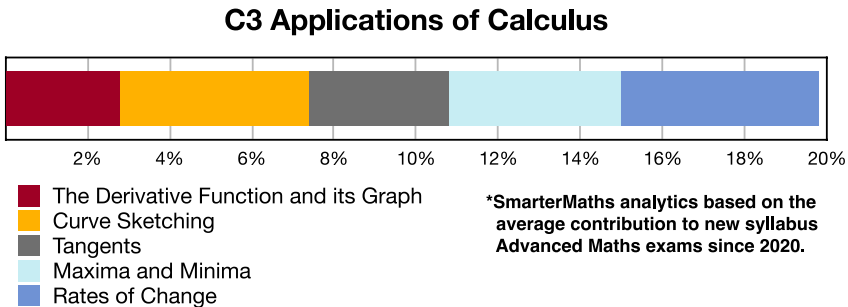
Stage 6

Calculus (Adv), C3 Applications of Calculus (Adv)

Curve Sketching (Y12)

Teacher: SHS

Exam Equivalent Time: 171 minutes (based on allocation of 1.5 minutes per mark)



HISTORICAL CONTRIBUTION

- C3 Applications of Calculus* is the 800 pound gorilla in the Advanced rainforest, contributing a massive 19.8% to new syllabus exams since introduced in 2020.
- This topic has been split into five sub-topics for analysis purposes: 1-*The Derivative Function and its Graph* (2.8%), 2-*Curve Sketching* (4.6%), 3-*Tangents* (3.4%), 4-*Maxima and Minima* (4.2%) and 5-*Rates of Change* (4.8%).
- This analysis looks at *Curve Sketching*.

HSC ANALYSIS - What to expect and common pitfalls

- Curve Sketching* (4.6%) is consistently examined with significant mark allocations.
- The most popular curve is easily the cubic, asked 6 times in the last decade, including each year in the period 2017-2020.
- Important: The 2022 Adv 22 cubic asked for the "global maximum and minimum" within a given range for the first time. This required finding the y -values of the graph at the domain limits - a must review question!
- Sketches of polynomials of degree 4 are the second most popular, most recently tested in 2024 (see 2024 Adv 19 and 2012 Adv 14a).
- Curve Sketching* questions regularly require students to look at concavity and intervals where a curve is increasing or decreasing. An important revision area that produces low mean marks.

Questions

1. Calculus, 2ADV C3 2011 HSC 7a

Let $f(x) = x^3 - 3x + 2$.

- Find the coordinates of the stationary points of $y = f(x)$, and determine their nature. (3 marks)
- Hence, sketch the graph $y = f(x)$ showing all stationary points and the y -intercept. (2 marks)

2. Calculus, 2ADV C3 2005 HSC 4b

A function $f(x)$ is defined by $f(x) = (x + 3)(x^2 - 9)$.

- Find all solutions of $f(x) = 0$ (2 marks)
- Find the coordinates of the turning points of the graph of $y = f(x)$, and determine their nature. (3 marks)
- Hence sketch the graph of $y = f(x)$, showing the turning points and the points where the curve meets the x -axis. (2 marks)
- For what values of x is the graph of $y = f(x)$ concave down? (1 mark)

3. Calculus, 2ADV C3 2006 HSC 5a

A function $f(x)$ is defined by $f(x) = 2x^2(3 - x)$.

- Find the coordinates of the turning points of $y = f(x)$ and determine their nature. (3 marks)
- Find the coordinates of the point of inflection. (1 mark)
- Hence sketch the graph of $y = f(x)$, showing the turning points, the point of inflection and the points where the curve meets the x -axis. (3 marks)
- What is the minimum value of $f(x)$ for $-1 \leq x \leq 4$? (1 mark)

4. Calculus, 2ADV C3 2007 HSC 6b

Let $f(x) = x^4 - 4x^3$.

- Find the coordinates of the points where the curve crosses the axes. (2 marks)
- Find the coordinates of the stationary points and determine their nature. (4 marks)
- Find the coordinates of the points of inflection. (1 mark)
- Sketch the graph of $y = f(x)$, indicating clearly the intercepts, stationary points and points of inflection. (3 marks)

5. Calculus, 2ADV C3 2015 HSC 13c

Consider the curve $y = x^3 - x^2 - x + 3$.

- Find the stationary points and determine their nature. (4 marks)
- Given that the point $P\left(\frac{1}{3}, \frac{70}{27}\right)$ lies on the curve, prove that there is a point of inflection at P . (2 marks)
- Sketch the curve, labelling the stationary points, point of inflection and y -intercept. (2 marks)

6. Calculus, 2ADV C3 2016 HSC 13a

Consider the function $y = 4x^3 - x^4$.

- i. Find the two stationary points and determine their nature. (4 marks)
- ii. Sketch the graph of the function, clearly showing the stationary points and the x and y intercepts. (2 marks)

7. Calculus, 2ADV C3 2017 HSC 13b

Consider the curve $y = 2x^3 + 3x^2 - 12x + 7$.

- i. Find the stationary points of the curve and determine their nature. (4 marks)
- ii. Sketch the curve, labelling the stationary points. (2 marks)
- iii. Hence, or otherwise, find the values of x for which $\frac{dy}{dx}$ is positive. (1 mark)

8. Calculus, 2ADV C3 2018 HSC 13a

Consider the curve $y = 6x^2 - x^3$.

- i. Find the stationary points and determine their nature. (3 marks)
- ii. Given that the point (2,16) lies on the curve, show that it is a point of inflection. (2 marks)
- iii. Sketch the curve, showing the stationary points, the point of inflection and the x and y intercepts. (2 marks)

9. Calculus, 2ADV C3 2020 HSC 16

Sketch the graph of the curve $y = -x^3 + 3x^2 - 1$, labelling the stationary points and point of inflection. Do NOT determine the x -intercepts of the curve. (4 marks)

10. Calculus, 2ADV C3 2022 HSC 27

Let $f(x) = xe^{-2x}$.

It is given that $f'(x) = e^{-2x} - 2xe^{-2x}$.

- a. Show that $f''(x) = 4(x - 1)e^{-2x}$. (2 marks)
- b. Find any stationary points of $f(x)$ and determine their nature. (2 marks)
- c. Sketch the curve $y = xe^{-2x}$, showing any stationary points, points of inflection and intercepts with the axes. (3 marks)

11. Calculus, 2ADV C3 2024 HSC 19

Sketch the curve $y = x^4 - 2x^3 + 2$ by first finding all stationary points, checking their nature, and finding the points of inflection. (5 marks)

12. Calculus, 2ADV C3 2019 HSC 14b

The derivative of a function $y = f(x)$ is given by $f'(x) = 3x^2 + 2x - 1$.

- a. Find the x -values of the two stationary points of $y = f(x)$, and determine the nature of the stationary points. (2 marks)
- b. The curve passes through the point (0, 4). Find an expression for $f(x)$. (2 marks)
- c. Hence sketch the curve, clearly indicating the stationary points. (2 marks)
- d. For what values of x is the curve concave down? (1 mark)

13. Calculus, 2ADV C3 2004 HSC 4b

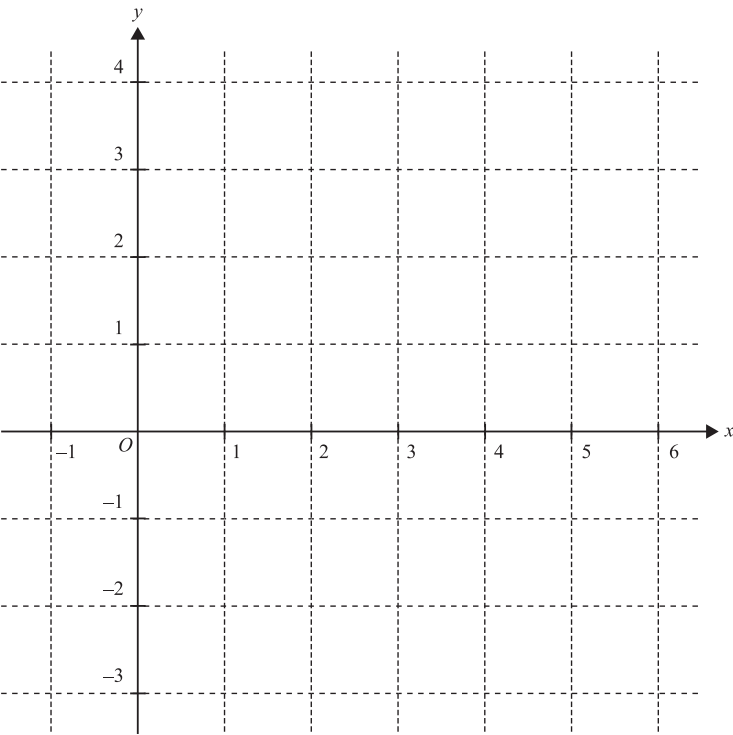
Consider the function $f(x) = x^3 - 3x^2$.

- i. Find the coordinates of the stationary points of the curve $y = f(x)$ and determine their nature. (3 marks)
- ii. Sketch the curve showing where it meets the axes. (2 marks)
- iii. Find the values of x for which the curve $y = f(x)$ is concave up. (2 marks)

14. Calculus, 2ADV C3 2019 MET1 4

Given the function $f(x) = \log_e(x - 3) + 2$,

- a. State the domain and range of $f(x)$. (1 mark)
- b. i. Find the equation of the tangent to the graph of $f(x)$ at $(4, 2)$. (2 marks)
- ii. On the axes below, sketch the graph of the function $f(x)$, labelling any asymptote with its equation. Also draw the tangent to the graph of $f(x)$ at $(4, 2)$. (4 marks)



15. Calculus, 2ADV C3 2010 HSC 6a

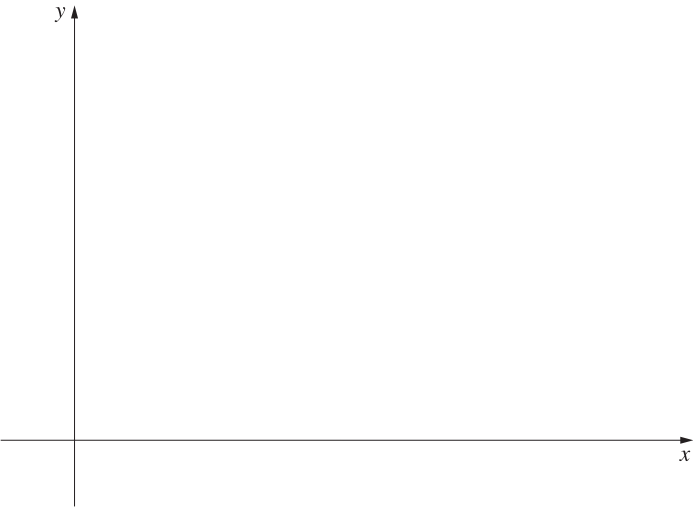
Let $f(x) = (x + 2)(x^2 + 4)$.

- i. Show that the graph $y = f(x)$ has no stationary points. (2 marks)
- ii. Find the values of x for which the graph $y = f(x)$ is concave down, and the values for which it is concave up. (2 marks)
- iii. Sketch the graph $y = f(x)$, indicating the values of the x and y intercepts. (2 marks)

16. Calculus, 2ADV C3 2023 HSC 30

Let $f(x) = e^{-x}\sin x$.

- a. Find the coordinates of the stationary points of $f(x)$ for $0 \leq x \leq 2\pi$. You do NOT need to check the nature of the stationary points. (3 marks)
- b. Without using any further calculus, sketch the graph of $y = f(x)$, for $0 \leq x \leq 2\pi$, showing stationary points and intercepts. (2 marks)



17. Calculus, 2ADV C3 2008 HSC 8a

Let $f(x) = x^4 - 8x^2$.

- i. Find the coordinates of the points where the graph of $y = f(x)$ crosses the axes. (2 marks)
- ii. Show that $f(x)$ is an even function. (1 mark)
- iii. Find the coordinates of the stationary points of $f(x)$ and determine their nature. (4 marks)
- iv. Sketch the graph of $y = f(x)$. (1 mark)

Worked Solutions

1. Calculus, 2ADV C3 2011 HSC 7a

i. $f(x) = x^3 - 3x + 2$
 $f'(x) = 3x^2 - 3$
 $f''(x) = 6x$

Stationary points when $f'(x) = 0$

$$3x^2 - 3 = 0$$

$$3(x^2 - 1) = 0$$

$$\therefore x^2 = 1$$

$$x = \pm 1$$

When $x = 1$

$$f(1) = 1 - 3 + 2 = 0$$

$$f''(1) = 6 > 0$$

$\therefore$ MIN S.P. at $(1, 0)$

When $x = -1$

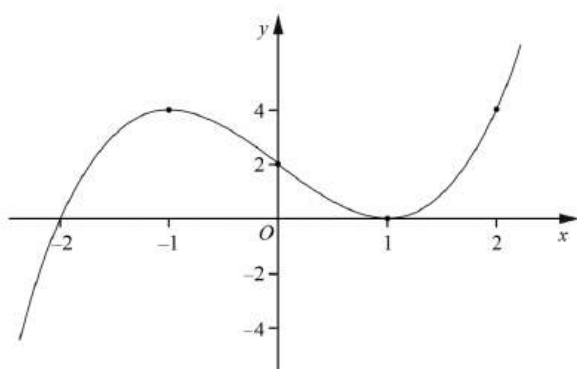
$$f(-1) = -1 + 3 + 2 = 4$$

$$f''(-1) = -6 < 0$$

$\therefore$ MAX S.P. at $(-1, 4)$

ii. $y = x^3 - 3x + 2$

y -intercept = 2



MARKER'S COMMENT: Graphs should be *large* (around $\frac{1}{2}$ page), axes drawn *with a ruler*, with intercepts and turning points clearly shown. The scale can be different on each axis for clarity, as shown in the Worked Solution.

2. Calculus, 2ADV C3 2005 HSC 4b

i. $f(x) = (x + 3)(x^2 - 9)$
 $= (x + 3)(x + 3)(x - 3)$
 $\therefore f(x) = 0$ when $x = -3$ or 3

ii. $f(x) = (x + 3)(x^2 - 9)$
 $= x^3 - 9x + 3x^2 - 27$
 $= x^3 + 3x^2 - 9x - 27$
 $f'(x) = 3x^2 + 6x - 9$
 $f''(x) = 6x + 6$

S.P.'s when $f'(x) = 0$

$$3x^2 + 6x - 9 = 0$$

$$3(x^2 + 2x - 3) = 0$$

$$3(x - 1)(x + 3) = 0$$

At $x = 1$

$$f(1) = (4)(-8) = -32$$

$$f''(1) = 6 + 6 = 12 > 0$$

$\therefore$ MIN at $(1, -32)$

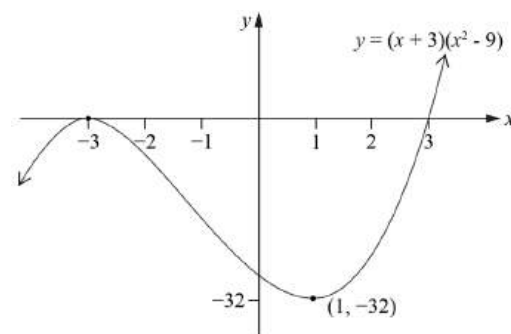
At $x = -3$

$$f(-3) = 0$$

$$f''(-3) = (6 \times -3) + 6 = -12 < 0$$

$\therefore$ MAX at $(-3, 0)$

iii.



iv. $f(x)$ is concave down when

$$f''(x) < 0$$

$$6x + 6 < 0$$

$$6x < -6$$

$$x < -1$$

3. Calculus, 2ADV C3 2006 HSC 5a

i. $f(x) = 2x^2(3-x)$

$$= 6x^2 - 2x^3$$

$$f'(x) = 12x - 6x^2$$

$$f''(x) = 12 - 12x$$

S.P.'s when $f'(x) = 0$

$$12x - 6x^2 = 0$$

$$6x(2-x) = 0$$

$$x = 0 \text{ or } 2$$

When $x = 0$

$$f(0) = 0$$

$$f''(0) = 12 - 0 = 12 > 0$$

$\therefore$ MIN at $(0, 0)$

When $x = 2$

$$f(2) = 2 \times 2^2(3-2) = 8$$

$$f''(2) = 12 - (12 \times 2) = -12 < 0$$

$\therefore$ MAX at $(2, 8)$

ii. P.I. when $f''(x) = 0$

$$12 - 12x = 0$$

$$12x = 12$$

$$x = 1$$

$$f''(0.5) = 6 > 0$$

$$f''(1.5) = -6 < 0$$

Since concavity changes $\Rightarrow$ P.I. exists

$$f(1) = 2 \times 1^2(3-1)$$

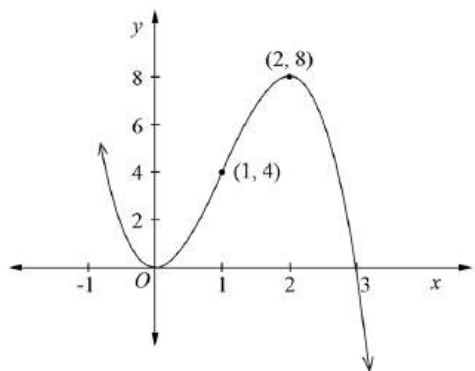
$$= 4$$

$\therefore$ P.I. at $(1, 4)$

iii. $f(x)$ meets x -axis when $f(x) = 0$

$$2x^2 \times (3-x) = 0$$

$$x = 0 \text{ or } 3$$



(iv) The graph clearly shows that in the given range $-1 \leq x \leq 4$, the minimum will occur when $x = 4$

$\therefore$ Minimum $= 2 \times 4^2(3-4)$

$= -32$

4. Calculus, 2ADV C3 2007 HSC 6b

i. $f(x) = x^4 - 4x^3$

Cuts x -axis when $f(x) = 0$

$$x^4 - 4x^3 = 0$$

$$x^3(x-4) = 0$$

$$x = 0 \text{ or } 4$$

$\therefore$ Cuts the x -axis at $(0, 0)$, $(4, 0)$

Cuts the y -axis when $x = 0$

$\therefore$ Cuts the y -axis at $(0, 0)$

ii. $f(x) = x^4 - 4x^3$

$$f'(x) = 4x^3 - 12x^2$$

$$f''(x) = 12x^2 - 24x$$

S.P.'s when $f'(x) = 0$

$$4x^3 - 12x^2 = 0$$

$$4x^2(x-3) = 0$$

$$x = 0 \text{ or } 3$$

When $x = 0$

$$f(0) = 0$$

$$f''(0) = 0$$

x	-0.5	0	0.5
$f''(x)$	> 0	0	< 0

Since concavity changes, a P.I. occurs at $(0, 0)$

When $x = 3$

$$f(3) = 3^4 - 4 \times 3^3$$

$$= -27$$

$$f''(3) = 12 \times 3^2 - 24 \times 3$$

$$= 36 > 0$$

$\therefore$ Minimum S.P. at $(3, -27)$

iii. P.I. when $f''(x) = 0$

$$12x^2 - 24x = 0$$

$$12x(x-2) = 0$$

$$x = 0 \text{ or } 2$$

P.I. at $(0, 0)$ (from(ii))

When $x = 2$

x	1.5	2	2.5
$f''(x)$	< 0	0	> 0

Since concavity changes, a P.I.

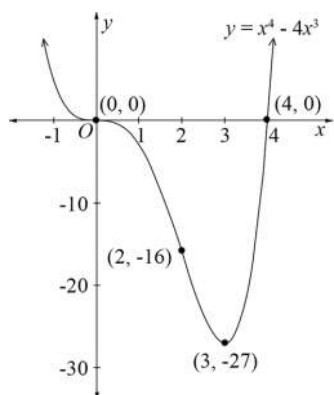
occurs when $x = 2$

$$f(2) = 2^4 - 4 \times 2^3$$

$$= 16$$

$\therefore$ P.I.'s at $(0, 0)$ and $(2, 16)$

iv.



5. Calculus, 2ADV C3 2015 HSC 13c

i. $y = x^3 - x^2 - x + 3$

$$\frac{dy}{dx} = 3x^2 - 2x - 1$$

$$\frac{d^2y}{dx^2} = 6x - 2$$

S.P.'s when $\frac{dy}{dx} = 0$

$$3x^2 - 2x - 1 = 0$$

$$(3x + 1)(x - 1) = 0$$

$$x = -\frac{1}{3} \text{ or } 1$$

When $x = -\frac{1}{3}$

$$f\left(-\frac{1}{3}\right) = \left(-\frac{1}{3}\right)^3 - \left(-\frac{1}{3}\right)^2 - \left(-\frac{1}{3}\right) + 3$$

$$= -\frac{1}{27} - \frac{1}{9} + \frac{1}{3} + 3$$

$$= \frac{86}{27}$$

$$f''\left(-\frac{1}{3}\right) = \left(6 \times -\frac{1}{3}\right) - 2 = -4 < 0$$

$$\therefore \text{MAX at } \left(-\frac{1}{3}, \frac{86}{27}\right)$$

When $x = 1$

$$f(1) = 1^3 - 1^2 - 1 + 3 = 2$$

$$f''(1) = (6 \times 1) - 2 = 4 > 0$$

$$\therefore \text{MIN at } (1, 2)$$

ii. $\frac{d^2y}{dx^2} = 0$ when

$$6x - 2 = 0$$

$$x = \frac{1}{3}$$

Checking change of concavity

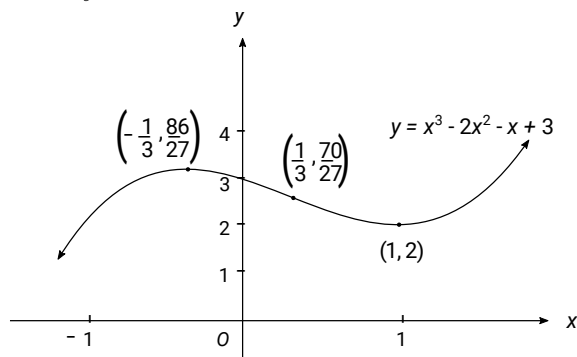
x	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$
$f(x)$	$-1 < 0$	0	$1 > 0$

Concavity changes either side of $x = \frac{1}{3}$

$\therefore \left(\frac{1}{3}, \frac{70}{27}\right)$ is a P.I.

iii. When $x = 0$

$$y = 3$$



6. Calculus, 2ADV C3 2016 HSC 13a

i.

$$y = 4x^3 - x^4$$

$$y' = 12x^2 - 4x^3$$

$$y'' = 24x - 12x^2$$

S.P.'s when $y' = 0$,

$$12x^2 - 4x^3 = 0$$

$$4x^2(3-x) = 0$$

$\therefore x = 0$ or 3

When $x = 0$, $y''(0) = 0$

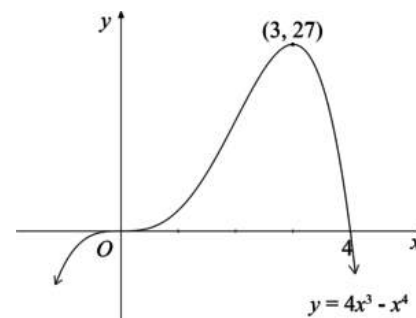
$\therefore$ P.I. at $(0, 0)$

When $x = 3$,

$$y''(3) = 24(3) - 12(9) = -36 < 0$$

$\therefore$ MAX at $(3, 27)$

ii.



7. Calculus, 2ADV C3 2017 HSC 13b

i. $y = 2x^3 + 3x^2 - 12x + 7$

$$\frac{dy}{dx} = 6x^2 + 6x - 12$$

$$\frac{d^2y}{dx^2} = 12x + 6$$

S.P. when $\frac{dy}{dx} = 0$

$$6x^2 + 6x - 12 = 0$$

$$x^2 + x - 2 = 0$$

$$(x + 2)(x - 1) = 0$$

$$x = -2 \text{ or } 1$$

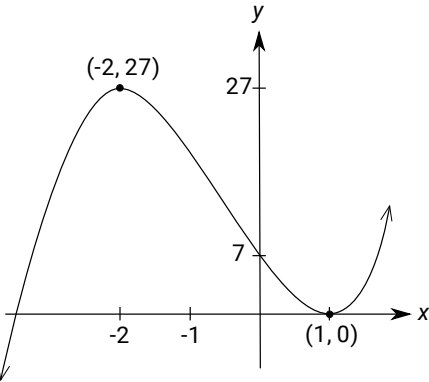
When $x = -2$, $\frac{d^2y}{dx^2} < 0$

$$\therefore \text{MAX at } (-2, 27)$$

When $x = 1$, $\frac{d^2y}{dx^2} > 0$

$$\therefore \text{MIN at } (1, 0)$$

ii.



iii. Solution 1

From graph, gradient is positive for

$$x < -2 \text{ and } x > 1$$

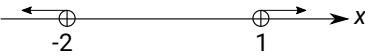
$$\therefore \frac{dy}{dx} > 0 \text{ for } x < -2 \text{ and } x > 1$$

Solution 2

$$\frac{dy}{dx} > 0$$

$$6x^2 + 6x - 12 > 0$$

$$(x + 2)(x - 1) > 0$$



$$\therefore x < -2 \text{ and } x > 1$$

8. Calculus, 2ADV C3 2018 HSC 13a

i. $y = 6x^2 - x^3$

$$\frac{dy}{dx} = 12x - 3x^2$$

$$\frac{d^2y}{dx^2} = 12 - 6x$$

S.P.s occur when $\frac{dy}{dx} = 0$

$$12x - 3x^2 = 0$$

$$3x(4 - x) = 0$$

$$x = 0 \text{ or } 4$$

When $x = 0$, $\frac{d^2y}{dx^2} > 0$

$\therefore$ MIN at $(0, 0)$

When $x = 4$, $\frac{d^2y}{dx^2} < 0$

$\therefore$ MAX at $(4, 32)$

ii. P.I. occur when $\frac{d^2y}{dx^2} = 0$,

$$12 - 6x = 0$$

$$x = 2$$

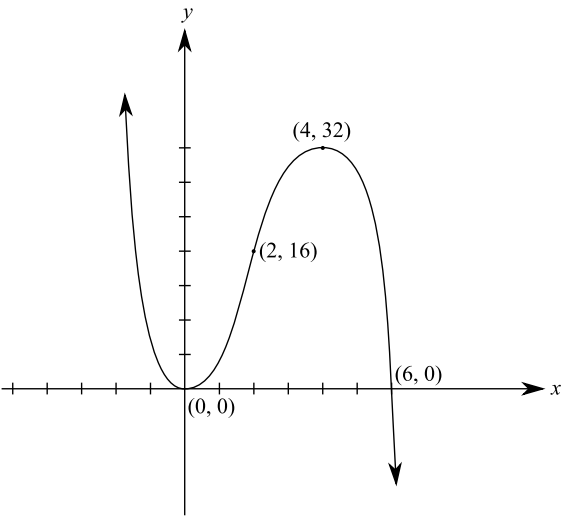
When $x = 2$, $y = 16$

x	2^-	2	2^+
$\frac{d^2y}{dx^2}$	> 0	0	< 0

Since the concavity changes

$\Rightarrow$ P.I. occurs at $(2, 16)$

iii.



9. Calculus, 2ADV C3 2020 HSC 16

$$y = -x^3 + 3x^2 - 1$$

$$\frac{dy}{dx} = -3x^2 + 6x$$

$$\frac{d^2y}{dx^2} = -6x + 6$$

SP's when $\frac{dy}{dx} = 0$

$$-3x^2 + 6x = 0$$

$$-3x(x-2) = 0$$
$$x = 0 \text{ or } 2$$

When $x = 0$,

$$y = -1$$

$$\frac{d^2y}{dx^2} = 6 > 0$$

∴ MIN at $(0, -1)$

When $x = 2$,

$$y = -8 + 12 - 1 = 3$$

$$\frac{d^2y}{dx^2} = -6 \times 2 + 6 = -6 < 0$$

∴ MAX at $(2, 3)$

$$\frac{d^2y}{dx^2} = 0 \text{ when}$$

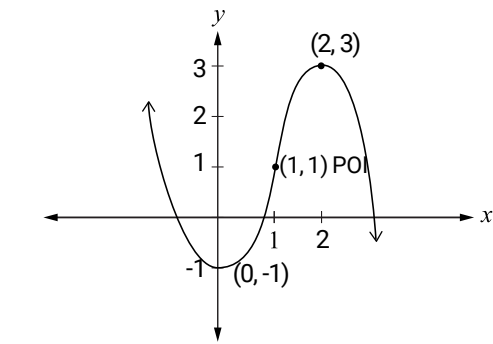
$$-6x + 6 = 0$$
$$x = 1$$

Checking change of concavity

x	$\frac{1}{2}$	1	$\frac{3}{2}$
$\frac{d^2y}{dx^2}$	3	0	-3

Concavity changes either side of $x = 1$

∴ POI at $(1, 1)$



10. Calculus, 2ADV C3 2022 HSC 27

a. $f'(x) = e^{-2x} - 2xe^{-2x}$

$$\begin{aligned} f''(x) &= -2e^{-2x} - [2x(-2)e^{-2x} + 2e^{-2x}] \\ &= -2e^{-2x} + 4xe^{-2x} - 2e^{-2x} \\ &= 4xe^{-2x} - 4e^{-2x} \\ &= 4(x-1)e^{-2x} \dots \text{as required} \end{aligned}$$

b. S.P.'s occur when $f'(x) = 0$

$$e^{-2x} - 2xe^{-2x} = 0$$

$$e^{-2x}(1 - 2x) = 0$$

$$e^{-2x} = 0 \Rightarrow \text{No solution}$$

$$1 - 2x = 0 \Rightarrow x = \frac{1}{2}$$

$$f''\left(\frac{1}{2}\right) = 4\left(\frac{1}{2} - 1\right)e^{-2 \times \frac{1}{2}}$$

$$= -2e^{-1} < 0 \Rightarrow \text{MAX}$$

$$f\left(\frac{1}{2}\right) = \frac{1}{2}e^{-1} = \frac{1}{2e}$$

c. POI occurs when $f''(x) = 0$

$$f''(x) = 4(x-1)e^{-2x} = 0 \Rightarrow x = 1$$

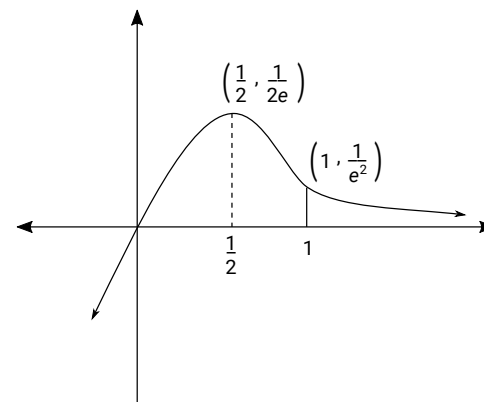
Test for change in concavity:

$$f''(0) = 4(0-1)e^0 = -4$$

$$f''(2) = 4(2-1)e^{-4} > 0 \Rightarrow \text{Concavity changes}$$

$$\therefore \text{ POI exists at } \left(1, \frac{1}{e^2}\right)$$

$$\text{As } x \rightarrow \infty \Rightarrow y \rightarrow 0^+$$



Mean mark part (c) 54%.

11. Calculus, 2ADV C3 2024 HSC 19

$y = x^4 - 2x^3 + 2$

$y' = 4x^3 - 6x^2 = 2x^2(2x - 3)$

$y'' = 12x^2 - 12x = 12x(x - 1)$

SP's when $y' = 0$:

$2x^2(2x - 3) = 0 \Rightarrow x = 0 \text{ or } \frac{3}{2}$

At $x = 0, y'' = 0$

x	-1	0	1
y'	-10	0	-5

$\Rightarrow$ Horizontal POI at $(0, 2)$

At $x = \frac{3}{2}$:

$y'' = 12 \times \frac{3}{2} \left(\frac{3}{2} - 1 \right) = 9 > 0, \quad y = \left(\frac{3}{2} \right)^4 - 2 \left(\frac{3}{2} \right)^3 + 2 = \frac{5}{16}$

$\Rightarrow$ MIN at $\left(\frac{3}{2}, \frac{5}{16} \right)$

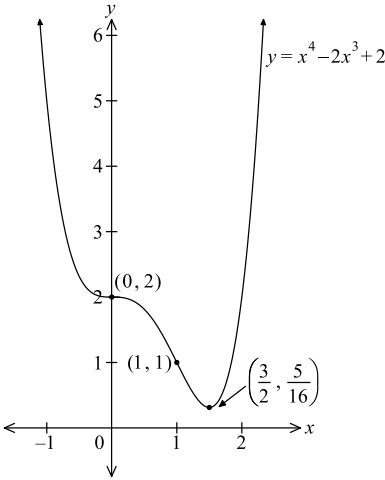
POI when $y'' = 0$:

$12x(x - 1) = 0 \Rightarrow x = 1 \text{ or } x = 0 \text{ (see above)}$

Test concavity change at $x = 1$:

x	$\frac{1}{2}$	1	$\frac{3}{2}$
y''	-3	0	9

$\Rightarrow$ POI at $(1, 1)$

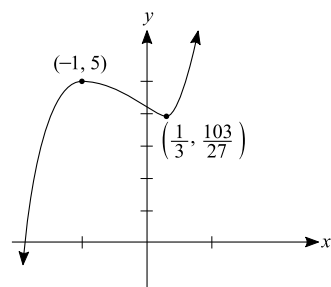


12. Calculus, 2ADV C3 2019 HSC 14b

- a. $f'(x) = 3x^2 + 2x - 1$
 $f''(x) = 6x + 2$
S.P.'s when $f'(x) = 0$
- $$3x^2 + 2x - 1 = 0$$
- $$(3x - 1)(x + 1) = 0$$
- $x = \frac{1}{3}$ or -1
- When $x = \frac{1}{3}$,
- $$f''(x) = 4 > 0 \Rightarrow \text{MIN}$$
- When $x = -1$,
- $$f''(x) = -4 < 0 \Rightarrow \text{MAX}$$

- b. $f(x) = \int f'(x) \, dx$
- $$= \int 3x^2 + 2x - 1 \, dx$$
- $$= x^3 + x^2 - x + c$$
- $(0, 4)$ lies on $f(x) \Rightarrow c = 4$
- $$\therefore f(x) = x^3 + x^2 - x + 4$$

- c. When $x = -1$, $y = 5$
- When $x = \frac{1}{3}$, $y = \frac{103}{27}$



- d. Concave down when $f''(x) < 0$

$$6x + 2 < 0$$
$$6x < -2$$
$$x$$

$$< -\frac{1}{3}$$

13. Calculus, 2ADV C3 2004 HSC 4b

(i) $f(x) = x^3 - 3x^2$

$$f'(x) = 3x^2 - 6x$$

$$f''(x) = 6x - 6$$

S.P.'s when $f'(x) = 0$

$$3x^2 - 6x = 0$$

$$3x(x-2) = 0$$

$$x = 0 \text{ or } 2$$

When $x = 0$

$$f(0) = 0$$

$$f''(0) = 0 - 6 = -6 < 0$$

$\therefore$ MAX at $(0, 0)$

When $x = 2$

$$f(2) = 2^3 - (3 \times 4) = -4$$

$$f''(2) = (6 \times 2) - 6 = 6 > 0$$

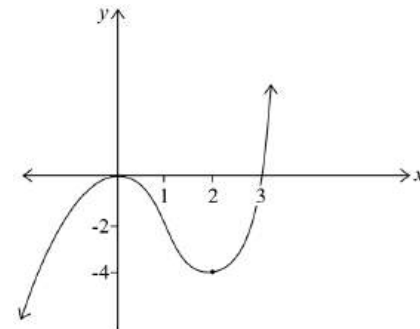
$\therefore$ MIN at $(2, -4)$

(ii) $f(x) = x^3 - 3x^2$ meets the x -axis when $f(x) = 0$

$$x^3 - 3x^2 = 0$$

$$x^2(x-3) = 0$$

$$x = 0 \text{ or } 3$$



(iii) $f(x)$ is concave up when

$$f''(x) > 0$$

$$6x - 6 > 0$$

$$6x > 6$$

$$x > 1$$

$\therefore f(x)$ is concave up when $x > 1$

14. Calculus, 2ADV C3 2019 MET1 4

a. Domain: $x > 3$

Range: $y \in R$

b.i. $g(x) = \log_e(x - 3) + 2$

$$g'(x) = \frac{1}{x - 3}$$

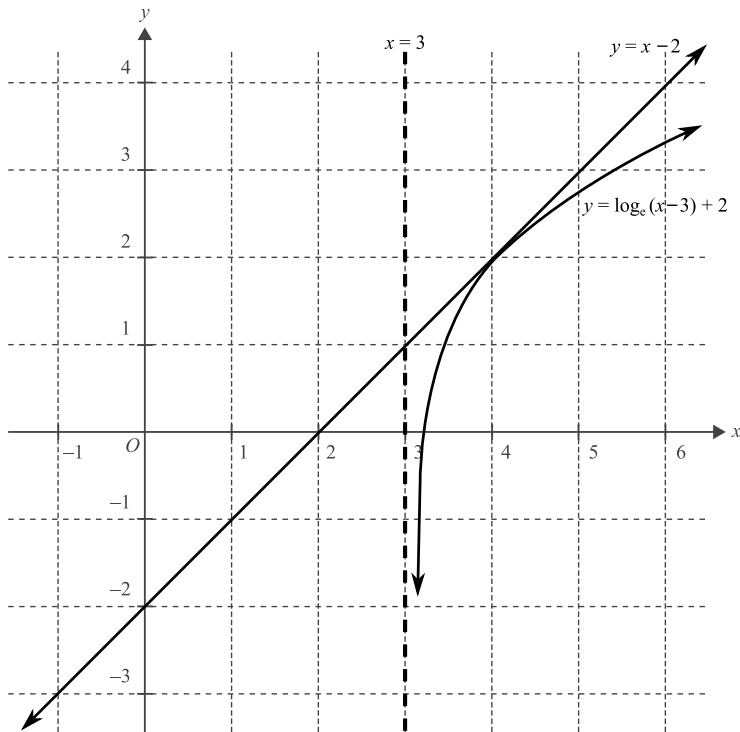
$$g'(4) = 1$$

Equation of tangent, $m = 1$ through $(4, 2)$:

$$y - 2 = 1(x - 4)$$

$$y = x - 2$$

b.ii.



15. Calculus, 2ADV C3 2010 HSC 6a

i. Need to show no S.P.'s

$$f(x) = (x + 2)(x^2 + 4)$$

$$= x^3 + 2x^2 + 4x + 8$$

$$f'(x) = 3x^2 + 4x + 4$$

S.P.s occur when $f'(x) = 0$,

$$3x^2 + 4x + 4 = 0$$

$$\Delta = b^2 - 4ac$$

$$= 4^2 - (4 \times 3 \times 4)$$

$$= 16 - 48$$

$$= -32 < 0$$

Since $\Delta < 0$, No Solution

$\therefore$ No S.P.'s for $f(x)$

ii. $f(x)$ is concave down when $f''(x) < 0$

$$f''(x) = 6x + 4$$

$$\Rightarrow 6x + 4 < 0$$

$$6x < -4$$

$$x < -\frac{2}{3}$$

$\therefore f(x)$ is concave down when $x < -\frac{2}{3}$

$f(x)$ is concave up when $f''(x) > 0$

$$f''(x) = 6x + 4$$

$$\Rightarrow 6x + 4 > 0$$

$$6x > -4$$

$$x > -\frac{2}{3}$$

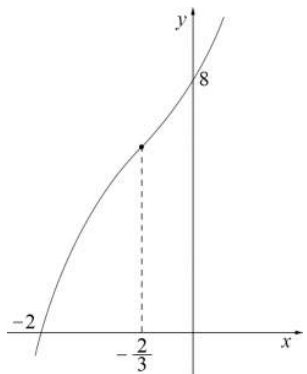
$\therefore f(x)$ is concave up when $x > -\frac{2}{3}$

iii. y -intercept $= 2 \times 4 = 8$

x -intercept $= -2$

MARKER'S COMMENT: The significance of the sign of the second derivative was not well understood by most students.

♦♦ Mean mark 33%.
MARKER'S COMMENT: Students are reminded to bring a ruler to the exam and use it to draw the axes for graphing and to help with an appropriate scale.



16. Calculus, 2ADV C3 2023 HSC 30

Mean mark (a) 54%.

a. $f(x) = e^{-x} \sin x$

$$f'(x) = e^{-x} \cos x - e^{-x} \sin x = e^{-x} (\cos x - \sin x)$$

SPs when $f'(x) = 0$:

$$e^{-x} = 0 \rightarrow \text{no solution}$$

$$\cos x - \sin x = 0$$

$$1 - \tan x = 0$$

$$\tan = 1$$

$$x = \frac{\pi}{4}, \frac{5\pi}{4}$$

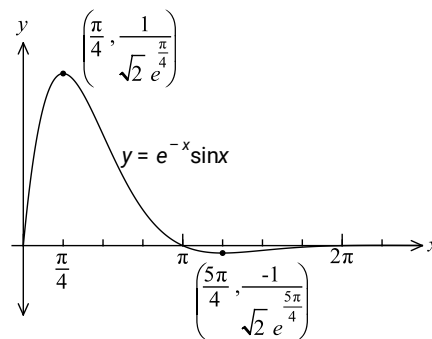
$$f\left(\frac{\pi}{4}\right) = e^{-\frac{\pi}{4}} \sin \frac{\pi}{4} = \frac{1}{\sqrt{2} \times e^{\frac{\pi}{4}}}$$

$$f\left(\frac{5\pi}{4}\right) = e^{-\frac{5\pi}{4}} \sin \frac{5\pi}{4} = \frac{-1}{\sqrt{2} \times e^{\frac{5\pi}{4}}}$$

$\therefore$ SPs at $\left(\frac{\pi}{4}, \frac{1}{\sqrt{2} \times e^{\frac{\pi}{4}}}\right)$ and $\left(\frac{5\pi}{4}, \frac{-1}{\sqrt{2} \times e^{\frac{5\pi}{4}}}\right)$

♦♦ Mean mark (b) 34%.

b. x -intercepts at $x = 0, \pi, 2\pi$



17. Calculus, 2ADV C3 2008 HSC 8a

i. $f(x) = x^4 - 8x^2$

y -intercept when $x = 0$

$\therefore$ Cuts y axis at $(0, 0)$

x -intercept when $f(x) = 0$

$$x^4 - 8x^2 = 0$$

$$x^2(x^2 - 8) = 0$$

$$x^2 - 8 = 0 \quad \text{or} \quad x = 0$$

$$x^2 = 8$$

$$x = \pm \sqrt{8} = \pm 2\sqrt{2}$$

$\therefore$ Cuts x -axis at $(-2\sqrt{2}, 0)$, and $(2\sqrt{2}, 0)$

(Note that it only touches x -axis at $(0, 0)$)

ii. $f(x) = x^4 - 8x^2$

$$f(-x) = (-x)^4 - 8(-x)^2$$

$$= x^4 - 8x^2$$

$$= f(x)$$

$\therefore f(x)$ is an even function.

iii. $f(x) = x^4 - 8x^2$

$$f'(x) = 4x^3 - 16x$$

$$f''(x) = 12x^2 - 16$$

S.P. when $f'(x) = 0$

$$4x^3 - 16x = 0$$

$$4x(x^2 - 4) = 0$$

$$x^2 - 4 = 0 \quad x = 0$$

$$x^2 = 4$$

$$x = \pm 2$$

At $x = 2$ $f(x) = (2)^4 - 8(2)^2 = -16$

$$f''(x) = 12(2^2) - 16 > 0$$

$\therefore$ MIN at $(2, -16)$

$\therefore$ MIN at $(-2, -16)$, ($f(x)$ is even)

At $x = 0$ $f(x) = (0)^4 - 8(0)^2 = 0$

$$f''(x) = 12(0^2) - 16 < 0$$

$\therefore$ MAX at $(0, 0)$

iv.

